

How to give read and write permissions using Unix commands.
A very simple example is to create a text file with a C program,
then use chmod.

Create a file called permission.c

```
#include <stdio.h>
int main()
{
    FILE *fp;
    fp = fopen("sample.txt", "w");
    if (fp == NULL)
    {
        printf("Cannot open file\n");
        return 1;
    }
    fprintf(fp, "Hello, this is a test file.\n");
    fclose(fp);
    printf("File created successfully.\n");
    return 0;
}
```

Compile

```
gcc permission.c -o permission
```

Run

```
./permission
```

Output

File created successfully.

A file called sample.txt is created.

Check the current permissions

```
ls -l sample.txt
```

You may see

```
-rw-r--r-- 1 user user 31 Aug 17 08:20 sample.txt
```

The owner has: Read + Write

Group has: Read

Others have: Read

This is permission 644.

Give Read and Write permission to everyone

```
chmod 666 sample.txt
```

Check `ls -l sample.txt`

You should see

```
-rw-rw-rw- 1 user user ... sample.txt
```

Instead of numbers, you can directly give:

`chmod u+rw sample.txt`

`chmod o+rw sample.txt`

`chmod a+rw sample.txt`

```
dell@DESKTOP-GM2LUJI:~/OSLab$ gedit permission.c
** (gedit:6482): WARNING **: 05:20:56.520: Could not load Peas repository: Typelib file for namespace 'Peas', version '1.0' not found
** (gedit:6482): WARNING **: 05:20:56.520: Could not load PeasGtk repository: Typelib file for namespace 'PeasGtk', version '1.0' not found
dell@DESKTOP-GM2LUJI:~/OSLab$ gcc permission.c -o permission
dell@DESKTOP-GM2LUJI:~/OSLab$ ./permission
File created successfully.
dell@DESKTOP-GM2LUJI:~/OSLab$ ls -l sample.txt
-rw-r--r-- 1 dell dell 28 Aug 17 05:21 sample.txt
dell@DESKTOP-GM2LUJI:~/OSLab$ chmod 666 sample.txt
dell@DESKTOP-GM2LUJI:~/OSLab$ ls -l sample.txt
-rw-rw-rw- 1 dell dell 28 Aug 17 05:21 sample.txt
dell@DESKTOP-GM2LUJI:~/OSLab$ chmod u+rw sample.txt
dell@DESKTOP-GM2LUJI:~/OSLab$ ls -l sample.txt
-rw-rw-rw- 1 dell dell 28 Aug 17 05:21 sample.txt
dell@DESKTOP-GM2LUJI:~/OSLab$ chmod o+rw sample.txt
dell@DESKTOP-GM2LUJI:~/OSLab$ ls -l sample.txt
-rw-rw-rw- 1 dell dell 28 Aug 17 05:21 sample.txt
dell@DESKTOP-GM2LUJI:~/OSLab$ chmod a+rw sample.txt
dell@DESKTOP-GM2LUJI:~/OSLab$ ls -l sample.txt
-rw-rw-rw- 1 dell dell 28 Aug 17 05:21 sample.txt
```

Task of the Day

- 1) Give Read and Write only to the owner
- 2) Give Read and Write to the owner and group
- 3) Only owner can read and write
- 4) Owner read/write, everyone else read-only
- 5) Owner can do everything, others can only read and execute
- 6) Owner and group can read/write
- 7) Only owner can execute

Outputs

```

dell@DESKTOP-GM2LUJI:~/OSLab$ chmod 600 sample.txt
dell@DESKTOP-GM2LUJI:~/OSLab$ ls -l sample.txt
-rw----- 1 dell dell 28 Aug 17 05:21 sample.txt
dell@DESKTOP-GM2LUJI:~/OSLab$ chmod 660 sample.txt
dell@DESKTOP-GM2LUJI:~/OSLab$ ls -l sample.txt
-rw-rw---- 1 dell dell 28 Aug 17 05:21 sample.txt
dell@DESKTOP-GM2LUJI:~/OSLab$ chmod 600 sample.txt
dell@DESKTOP-GM2LUJI:~/OSLab$ ls -l sample.txt
-rw----- 1 dell dell 28 Aug 17 05:21 sample.txt
dell@DESKTOP-GM2LUJI:~/OSLab$ chmod 644 sample.txt
dell@DESKTOP-GM2LUJI:~/OSLab$ ls -l sample.txt
-rw-r--r-- 1 dell dell 28 Aug 17 05:21 sample.txt
dell@DESKTOP-GM2LUJI:~/OSLab$ chmod 755 sample.txt
dell@DESKTOP-GM2LUJI:~/OSLab$ ls -l sample.txt
-rwxr-xr-x 1 dell dell 28 Aug 17 05:21 sample.txt
dell@DESKTOP-GM2LUJI:~/OSLab$ chmod 660 sample.txt
dell@DESKTOP-GM2LUJI:~/OSLab$ ls -l sample.txt
-rw-rw---- 1 dell dell 28 Aug 17 05:21 sample.txt
dell@DESKTOP-GM2LUJI:~/OSLab$ chmod 100 sample.txt
dell@DESKTOP-GM2LUJI:~/OSLab$ ls -l sample.txt
---x----- 1 dell dell 28 Aug 17 05:21 sample.txt
dell@DESKTOP-GM2LUJI:~/OSLab$

```

Fill in the permission numbers for the following:

8) Owner: read + write; Group: nothing; Others: nothing
→600

9) Owner: read + write; Group: read; Others: read →644

10) Owner: read + write + execute; Group: read + execute;
Others: read + execute → 755

11) Owner: read + write; Group: read + write; Others:
nothing →660

12) Everyone: read only →444

```

alekhya@Alekhya: ~
alekhya@Alekhya:~$ gedit permission1.c&
[1] 497
alekhya@Alekhya:~$
** (gedit:497): WARNING **: 03:10:59.704: Could not load Peas repository: Typelib file for namespace 'Peas', version '1.0' not found

** (gedit:497): WARNING **: 03:10:59.704: Could not load PeasGtk repository: Typelib file for namespace 'PeasGtk', version '1.0' not found
^C
[1]+  Done                  gedit permission1.c
alekhya@Alekhya:~$ gcc permission1.c -o permssion1
alekhya@Alekhya:~$ ./permission1
-bash: ./permission1: No such file or directory
alekhya@Alekhya:~$ ./permssion1
File created successfully.
alekhya@Alekhya:~$ ls
OS-KLU  OSLab  OScProgram  backspace.c  examplefork.c  fork11.c  hello.c  permission1.c  process  sample.txt
OSLAB  OSSP   backspace  examplefork  fork11      hello    helloshell  permssion1    process.c
alekhya@Alekhya:~$ ls -l sample.txt
-rw-r--r-- 1 alekhya alekhya 28 Aug 17 03:11 sample.txt
alekhya@Alekhya:~$ chmod 666 sample.txt
alekhya@Alekhya:~$ ls -l sample.txt
-rw-rw-rw- 1 alekhya alekhya 28 Aug 17 03:11 sample.txt
alekhya@Alekhya:~$ chmod 600 sample.txt
alekhya@Alekhya:~$ ls -l sample.txt
-rw----- 1 alekhya alekhya 28 Aug 17 03:11 sample.txt
alekhya@Alekhya:~$ chmod 660 sample.txt
alekhya@Alekhya:~$ ls -l sample.txt
-rw-rw---- 1 alekhya alekhya 28 Aug 17 03:11 sample.txt
alekhya@Alekhya:~$

```